

Benchmarking & Methodology (Slides 7–9)

"Hi everyone, I'll be walking you through our Benchmarking and Experimental Methodology.

To tackle this classification challenge, we selected three models with increasing levels of complexity. We started with Logistic Regression and SVM as our linear baselines, relying on TF-IDF features. Our core hypothesis was that RoBERTa, being a Transformer-based model, would significantly outperform the baselines. This is because RoBERTa understands the context and sequence of words, rather than just looking at keyword frequency."

Data Preprocessing (Slide 11)

"Before feeding data into these models, we performed standard preprocessing. However, we had one counter-intuitive finding regarding Stop Words.

While standard NLP practices often remove them, we found that keeping stop words was actually more effective for this specific task. It turns out that common words like 'the' or 'is' help maintain the natural sentence structure, which is vital for the model to detect the urgency and tone of a disaster tweet."

Logistic Regression Tuning (Slide 12)

"Moving on to our first experiment: Logistic Regression Tuning.

The key takeaway here is the importance of handling class imbalance. As you can see from our results, simply setting the Class Weight to 'Balanced' boosted our accuracy by 4%. Combined with a min_df of 2 to capture rarer terms, we achieved our best baseline accuracy of 76%."

SVM Tuning (Slide 13)

"Next, we tuned the Support Vector Machine.

Here, the Regularization parameter (C) was the deciding factor. We found that a smaller C value of 0.5 was optimal, reaching 77% accuracy. Interestingly, when we increased C to 2.0, the model tried too hard to fit the noisy outliers, which actually led to overfitting and a drop in performance."

RoBERTa Results & Analysis (Slide 14)

"Finally, let's look at our top performer: RoBERTa.

Training this large model required specific optimizations. First, we used FP16 Mixed Precision, which allowed us to train 2x faster on the GPU without losing accuracy. Second, we applied a very small Learning Rate of 2e-5. Since RoBERTa is already pre-trained, we only wanted to 'nudge' the weights gently, rather than pushing them too hard and destroying its prior knowledge.

The result? We achieved a Macro F1 score of nearly 90%. This metric is crucial because a high F1 score implies high Recall. In disaster scenarios, missing a distress call—a False Negative—is unacceptable. Our model is specifically tuned to prioritize catching every potential risk, ensuring we don't miss life-threatening events."